

This assignment will not be graded and is only for practice.

1) Let L_1 and L_2 be affine subspaces of $\mathbb{R}^3$, where $L_1 = U$ and $L_2 = (2, 0, 1) + W$, for some **1 point** vector subspaces U and W of $\mathbb{R}^3$. Let a basis for U be given by $\{(2, 0, 1), (1, 1, 0), (0, 1, 0)\}$ and a basis for W be given by $\{(1, 0, 1), (0, 1, 1)\}$. Suppose there is a linear transformation $T : U \rightarrow W$ such that $(0, 1, 0) \in \ker(T)$, $T(2, 0, 1) = (0, 1, 1)$ and $T(1, 1, 0) = (1, 0, 1)$. An affine mapping $f : L_1 \rightarrow L_2$ is obtained by defining $f(u) = (2, 0, 1) + T(u)$, for all $u \in U$. Which of the following options are true?

- ☐ $L_1 = \mathbb{R}^3$
- ☐ $L_2 = \{(x, y, z) \mid x + y - z = 1\}$
- ☐ $L_2 = \mathbb{R}^3$
- ☐ $f(x, y, z) = (x - 2z + 2, z, x - z + 1)$
- ☐ $f(x, y, z) = (x - 2z, z, x - z)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$L_1 = \mathbb{R}^3$$

$$L_2 = \{(x, y, z) \mid x + y - z = 1\}$$

$$f(x, y, z) = (x - 2z + 2, z, x - z + 1)$$

Solution:

Recall the definition of Affine map:

Let L_1 and L_2 be affine subspaces of V_1 and V_2 , respectively, and let U and W be the unique subspaces that correspond to L_1 and L_2 , respectively.

- A map $f : L_1 \rightarrow L_2$ is an affine map if there exists a linear transformation $T : U \rightarrow W$ such that

$$f(v_1 + u) = v_2 + T(u),$$

where v_1 and v_2 are elements of V_1 and V_2 , respectively such that $L_1 = v_1 + U$ and $L_2 = v_2 + W$.

Before we proceed to the solution you may want to recall the following weekly contents:

- Watch : Week 7: Lecture 2.
- Solve : Activity 7.2: Questions 4, 5 and 7.

Discussion on Option 1: It is given that $L_1 = U$ and $\{(2, 0, 1), (1, 1, 0), (0, 1, 0)\}$ is a basis of the vector subspace U .

2) The dimension of U is

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 3

1 point

$L_1 = U$ is a subspace of $\mathbb{R}^3$. To know whether $L_1 = U$ is a proper subspace of $\mathbb{R}^3$ or not, we just have to compare their dimensions.

3) Now, can you see why $L_1 = U = \mathbb{R}^3$?

1 point

☐ Yes

☐ No

No, the answer is incorrect.

Score: 0

Feedback:

Feedback: Dimension of a proper subspace of a vector space is always less than the dimension of the vector space. Here, $\dim(L_1) = \dim(\mathbb{R}^3)$.

Accepted Answers:

Yes

Discussion on Options 2 and 3: It is given that $L_2 = (2, 0, 1) + W$ and $\{(1, 0, 1), (0, 1, 1)\}$ is a basis for the vector subspace W . Now to find the explicit description of L_2 , we have to find the explicit description of W first.

$$\begin{aligned} W &= \{a(1, 0, 1) + b(0, 1, 1) \mid a, b \in \mathbb{R}\} \\ &= \{(a, b, a+b) \mid a, b \in \mathbb{R}\} \end{aligned}$$

As W is a subspace of $\mathbb{R}^3$, we want to express it using the usual x, y, z co-ordinate and the relation between them. From the above description we have, $x = a, y = b, z = a + b$, which implies that $z = x + y$. So

$$W = \{(x, y, z) \mid x + y - z = 0, \text{ where } x, y, z \in \mathbb{R}\}$$

4) Now, are you able to find the equation of L_2 ?

1 point

☐ Yes

☐ No

No, the answer is incorrect.

Score: 0

Feedback:

$$L_2 = (2, 0, 1) + W = (2, 0, 1) + \{(x, y, z) \mid x + y - z = 0\} = \{(2+x, y, z+1) \mid x + y - z = 0\}$$

Accepted Answers:

Yes

5) Do you agree that the sets $\{(2+x, y, z+1) \mid x+y-z=0\}$ and $\{(a, b, c) \mid a+b-c=1\}$ **1 point** are the same?

☐ Yes

☐ No

No, the answer is incorrect.

Score: 0

Feedback:

Feedback:

$$a = 2 + x, \quad b = y, \quad c = z + 1 \quad \text{such that } x + y - z = 0.$$

$$a + b - c = 2 + x + y - z - 1 = x + y - z + 1 = 0 + 1.$$

Accepted Answers:

Yes

Discussion on Option 4 and 5:

Step 1: Here, we have to find an algebraic expression for the affine map $f : L_1 \rightarrow L_2$. To get an algebraic expression for f , we have to find the algebraic expression which represents the linear transformation $T : U \rightarrow W$.

Given:

$$T(0, 1, 0) = (0, 0, 0), \quad T(2, 0, 1) = (0, 1, 1). \quad T(1, 1, 0) = (1, 0, 1)$$

6) Is T uniquely determined by the above information?

1 point

☐ Yes

☐ No

No, the answer is incorrect.

Score: 0

Feedback:

Feedback: As $\{(0, 1, 0), (2, 0, 1), (1, 1, 0)\}$ forms a basis of U and T is defined for each element of this basis, then the above description completely and uniquely describes T .

Accepted Answers:

Yes

Step 2: Since $\{(0, 1, 0), (2, 0, 1), (1, 1, 0)\}$ is a basis of $U = \mathbb{R}^3$, we can write any element $(x, y, z) \in U$ as a linear combination of $\{(0, 1, 0), (2, 0, 1), (1, 1, 0)\}$.

$$(1) \quad (x, y, z) = a(0, 1, 0) + b(2, 0, 1) + c(1, 1, 0)$$

Apply T on both sides:

$$T(x, y, z) = aT(0, 1, 0) + bT(2, 0, 1) + cT(1, 1, 0)$$

$$T(x, y, z) = a(0, 0, 0) + b(0, 1, 1) + c(1, 0, 1)$$

To get an algebraic expression for $T(x, y, z)$, we just have to know the values of a , b and c in terms of x, y and z .

7) Now, can you find the values of a , b and c using equation (1)?

1 point

☐ Yes

☐ No

No, the answer is incorrect.

Score: 0

Feedback:

Feedback: Form equation (1), we get

$$2b + c = x$$

$$a + c = y$$

$$b = z$$

Accepted Answers:

Yes

Solve the above system for a , b and c .

8) What is the value of a ?

1 point

- ☐ z
- ☐ $x - 2z$
- ☐ $y + 2z - x$
- ☐ $x - y$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$y + 2z - x$

9) What is the value of b ?

1 point

- ☐ z
- ☐ $x - 2z$
- ☐ $y + 2z - x$
- ☐ $x - y$

No, the answer is incorrect.

Score: 0

Accepted Answers:

z

10) What is the value of c ?

1 point

- ☐ z
- ☐ $x - 2z$
- ☐ $y + 2z - x$
- ☐ $x - y$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$x - 2z$

Step 3:

$$T(x, y, z) = (0, z, z) + (x - 2z, 0, x - 2z) = (x - 2z, z, x - z)$$

By definition:

$$f(x, y, z) = (2, 0, 1) + T(x, y, z) \quad \text{where } (x, y, z) \in U.$$

By substituting the value $T(x, y, z)$ in the above expression, we get

$$f(x, y, z) = (x - 2z + 2, z, x - z + 1) \quad \text{for all } (x, y, z) \in U.$$

Your score is: 0/10